

VIETNAM NATIONAL UNIVERSITY HANOI
UNIVERSITY OF ENGINEERING AND TECHNOLOGY



PHAM MAI ANH

AN AI-DRIVEN CLOSED-LOOP FRAMEWORK
FOR REAL-TIME DDOS DETECTION
AND MITIGATION IN DATA CENTERS

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Major: Computer Science

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Supervisor:

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My thesis still has potential for enhancement, I welcome any feedback that could help improve the quality of this thesis in the future.

Sincere gratitude!

Authorship

I declare that the thesis is my original work, under the supervision of Associate Professor Nguyen Ngoc Hoa and Doctor Du Phuong Hanh. This work has not been submitted for any other degree or professional qualification.

All references and related works used in this thesis have been appropriately acknowledged and cited. The statistical data included has been thoroughly tested and analyzed based on actual experimental results.

I affirm that this declaration is made with full integrity and in accordance with the institution's policies.

Author

Pham Mai Anh

AN AI-DRIVEN CLOSED-LOOP FRAMEWORK FOR REAL-TIME DDoS DETECTION AND MITIGATION IN DATA CENTERS

Pham Mai Anh

Abstract

DDoS attacks remain a critical threat to modern data centers, causing service disruption, resource exhaustion, and significant operational losses. Traditional mitigation approaches, often based on static rule-based systems, lack the flexibility and scalability required to respond to increasingly dynamic and large-scale attack patterns. Recent advances in SDN, NFV, and AI offer new opportunities for building adaptive and automated defense mechanisms.

This thesis proposes CLAD, a Closed-loop Learning-based Adaptive DDoS Defense framework for real-time DDoS detection and mitigation in cloud-native data center environments. The framework integrates flow-level traffic monitoring with machine learning-based detection models, including machine learning and deep learning techniques, to identify anomalous network behavior. To enhance detection reliability, a validation module is incorporated to filter false positives using threshold-based and rule-based verification mechanisms.

Upon confirming an attack, the system generates high-level intent-based security policies, which are translated into enforceable actions through a policy management module. A dynamic allocation engine is then employed to determine the optimal deployment of VNFs, such as firewalls, intrusion detection systems, and rate limiters, based on the attack characteristics and system conditions. The SDN controller enforces traffic redirection by installing appropriate flow rules, enabling effective mitigation while preserving normal service operations.

The proposed closed-loop architecture incorporates continuous monitoring and feedback, allowing the system to adapt to evolving attack patterns and improve its detection and mitigation performance over time. Experimental evaluation demonstrates that the framework achieves high detection accuracy and efficient real-time response with minimal impact on legitimate traffic.

Overall, this work presents a scalable and adaptive solution that combines AI-driven detection, intent-based orchestration, and NFV-enabled mitigation, contributing to enhanced resilience against DDoS attacks in modern data center environments.

Keywords: *DDoS defense, machine learning, closed-loop automation, SDN/NFV, CNF-based mitigation, cloud-native security*

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List of Abbreviations

DDoS	Distributed Denial of Service
SDN	Software-Defined Networking
NFV	Network Function Virtualization
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
VNFs	Virtual Network Functions
API	Application Programming Interface
ONOS	Open Networking Operating System
ESTI	European Telecommunications Standards Institute
MANO	Management and Orchestration
NFVI	Network Functions Virtualization Infrastructure
NFVO	Network Functions Virtualization Orchestrator
VNFM	Virtual Network Function Manager
VIM	Virtual Infrastructure Manager
ONAP	Open Network Automation Platform
CNF(s)	Containerized Network Functions
VES	Virtual Event Streaming
SGD	Stochastic Gradient Descent
XGB	eXtreme Gradient Boosting
LightGBM	Light Gradient Boosting Machine
CNN	Convolutional Neural Network
SLAs	Service-Level Agreements
SVM	Support Vector Machine
KNN	K-Nearest Neighbors
DMaaP	Data Movement as a Platform
DCAE	Data Collection Analytics and Events

Introduction

Motivation

The rapid growth of cloud computing, data center infrastructures, and large-scale Internet services has significantly increased the demand for high-performance and reliable networking. Modern data center networks must handle massive volumes of traffic while maintaining low latency and high availability. In such environments, DDoS attacks pose a critical threat, as they can disrupt services by overwhelming network and system resources, leading to severe performance degradation or complete service outages.

The increasing scale and sophistication of DDoS attacks, driven by the proliferation of botnets and compromised devices, have exposed the limitations of traditional defense mechanisms. Conventional approaches, including rule-based filtering and signature-based intrusion detection systems, often struggle to cope with dynamic attack behaviors and previously unseen threats. As a result, there is a growing need for intelligent security solutions capable of adapting to evolving attack patterns and responding in real time.

Recent advances in AI, particularly ML and DL, have created new opportunities for improving DDoS detection. By learning traffic characteristics directly from data, AI-based approaches can identify both known and unknown attacks with higher accuracy than conventional methods. At the same time, programmable networking and virtualized network infrastructures provide the flexibility required to deploy security functions dynamically and automate mitigation processes.

Despite these advances, many existing solutions primarily focus on improving detection performance while treating mitigation as a separate process. This separation introduces delays between attack identification and response execution, reducing the overall effectiveness of defense mechanisms in fast-changing environments. Furthermore, limited integration between analytics, decision-making, and mitigation components restricts the ability of security systems to adapt continuously to evolving attack conditions.

These limitations motivate the development of an AI-driven closed-loop framework that tightly integrates detection, decision-making, and mitigation within a unified workflow. By combining intelligent traffic analysis with automated orchestration and adaptive response mechanisms, such a framework can provide a more scalable, responsive, and resilient defense against modern DDoS threats in data center environments.

Research Challenges and Problem Statement

Despite recent advancements in network security, defending against DDoS attacks in modern data center environments remains a challenging task. The increasing scale, diversity, and sophistication of attack strategies introduce several critical challenges that existing solutions have not fully addressed.

1. **Accurate and efficient DDoS detection.** The increasing diversity and sophistication of DDoS attacks make reliable detection a challenging task. Attack traffic patterns continuously evolve and may closely resemble legitimate network behavior, reducing the effectiveness of traditional signature-based approaches. Although AI-based methods have significantly improved detection capability, achieving high accuracy while maintaining low computational overhead and supporting real-time operation remains a critical challenge.
2. **Real-time and adaptive mitigation.** Timely mitigation is essential for minimizing the impact of DDoS attacks on service availability. However, applying effective countermeasures without disrupting legitimate traffic remains difficult, particularly in large-scale and dynamic environments. Mitigation strategies must not only respond rapidly to detected attacks but also adapt to changing attack characteristics and network conditions.
3. **Closed-loop integration of detection and mitigation** Many existing DDoS defense solutions focus primarily on either detection or mitigation, treating them as separate processes. This separation often introduces delays between attack identification and response execution, limiting the system's ability to react effectively to evolving threats. Developing a unified framework that continuously connects detection outcomes with mitigation actions through a feedback-driven process remains a significant challenge for building intelligent and adaptive defense systems.

Motivated by these challenges, this thesis investigates how to design an AI-driven closed-loop framework that enables accurate real-time DDoS detection and adaptive mitigation in modern data center environments while maintaining scalability, efficiency, and minimal impact on legitimate traffic.

To address this problem, the proposed framework integrates AI-based traffic analysis with feedback-driven mitigation mechanisms, enabling continuous interaction between detection and response components within a unified defense workflow.

Research Objective and Tasks

Motivated by the challenges of achieving accurate DDoS detection, enabling real-time mitigation, and effectively integrating these mechanisms within SDN/NFV-enabled environments, this thesis focuses on the design of an intelligent and adaptive framework for DDoS

defense. The proposed approach enhances detection capability by leveraging machine learning techniques to identify both known and previously unseen attack patterns under dynamic network conditions.

In addition to improving detection performance, this work develops efficient mitigation strategies that respond promptly to detected attacks while minimizing the impact on legitimate traffic. Particular attention is given to ensuring that mitigation actions remain effective without degrading overall network performance in high-speed environments.

Furthermore, this thesis integrates detection and mitigation into a unified, closed-loop framework to enable faster and more adaptive responses to evolving attack scenarios. The programmability of SDN and the flexibility of NFV are leveraged to support dynamic and scalable deployment of security functions.

To accomplish these objectives, the thesis is organized around the following tasks:

- Designing an SDN/NFV-based architecture for DDoS detection and mitigation.
- Developing a machine learning-based model for accurate and adaptive DDoS detection.
- Implementing real-time mitigation strategies to handle detected attacks effectively.
- Integrating detection and mitigation into a closed-loop framework for automated response.
- Evaluating the proposed framework using benchmark datasets and performance metrics.

Contribution Highlights

This thesis proposes CLAD, a Closed-loop Learning-based Adaptive DDoS Defense framework that integrates flow-level traffic monitoring, learning-based detection, intent-based policy decision, CNF-based mitigation, and feedback-driven adaptation into a unified defense workflow. This design improves both detection effectiveness and response flexibility in dynamic network environments. The key contributions of this work are summarized as follows:

1. **AI-driven detection module**, which investigates multiple ML and DL models, including SGD, XGB, LightGBM, CatBoost, and CNN, for identifying malicious traffic patterns. In addition, an ensemble voting strategy combining XGB, LightGBM, and CatBoost is explored to leverage the complementary strengths of gradient-boosting models and enhance overall detection performance.
2. **Adaptive mitigation module**, which transforms detection outcomes into high-level mitigation decisions and subsequently applies appropriate countermeasures through automated security policies. This mechanism enables timely and context-aware responses while minimizing the impact on legitimate traffic.

Together, these components provide an intelligent and adaptive defense mechanism that tightly couples detection and mitigation within a unified framework.

Thesis Outline

This thesis is organized into four main chapters.

- Chapter 1 introduces the background concepts relevant to this study, including DDoS defense, AI-based detection, and modern network security infrastructures.
- Chapter 2 reviews related work on existing AI-based DDoS detection and mitigation approaches, analyzes their limitations, and identifies the research gaps addressed in this thesis.
- Chapter 3 presents the proposed AI-driven closed-loop framework, including its architecture, detection mechanism, mitigation process, and feedback-driven operation.
- Chapter 4 describes the implementation, evaluation setup, and results, demonstrating detection performance and response efficiency.

The thesis concludes with a summary of findings and directions for future research.

Chapter 1

Background

Modern data centers are increasingly exposed to large-scale and sophisticated DDoS attacks, which threaten service availability and network reliability. Addressing these threats requires not only accurate attack detection but also timely and adaptive mitigation mechanisms. Recent advances in AI have significantly improved traffic analysis and anomaly detection capabilities, while technologies such as SDN, NFV, Kubernetes, and ONAP provide flexible platforms for deploying and orchestrating security functions. This chapter introduces the fundamental concepts and enabling technologies that form the foundation of the proposed AI-driven closed-loop framework for real-time DDoS detection and mitigation.

1.1 DDoS Threats in Modern Data Centers

DDoS attacks are among the most significant security threats facing modern data centers. By overwhelming network links, servers, or application resources with massive volumes of malicious traffic, these attacks can severely degrade service performance or render critical services unavailable. The increasing availability of botnets and compromised Internet-connected devices has enabled attackers to launch large-scale attacks with unprecedented traffic volumes and complexity.

Unlike traditional network environments, modern data centers host a large number of cloud services, virtualized resources, and containerized applications that must remain continuously accessible to users. Consequently, service disruptions caused by DDoS attacks can lead to financial losses, degraded user experience, and violations of SLAs. Furthermore, the diversity of attack vectors, ranging from volumetric flooding attacks to protocol and application-layer attacks, makes defense increasingly challenging.

Another important challenge is the requirement for rapid response. DDoS attacks can escalate within a short period of time, leaving limited opportunities for manual intervention. Conventional rule-based security mechanisms often struggle to adapt to evolving attack patterns and may generate delayed or ineffective responses. As a result, there is a growing need for intelligent defense systems capable of accurately detecting malicious traffic and automatically initiating mitigation actions in real time.

These challenges motivate the adoption of AI-driven detection techniques and adaptive mitigation mechanisms, which together form the foundation of the closed-loop framework proposed in this thesis.

1.2 AI-Based DDoS Detection

To overcome the limitations of traditional rule-based approaches, machine learning techniques have been widely adopted for DDoS detection. These methods enable automatic learning of traffic patterns and effective discrimination between normal and malicious behaviors.

This thesis investigates several machine learning and deep learning techniques for DDoS detection, including SGD, XGB, LightGBM, CatBoost, and CNN.

- **Stochastic Gradient Descent.** SGD is a lightweight and computationally efficient learning approach suitable for large-scale and streaming data. Its fast training and low latency make it appropriate for real-time DDoS detection.
- **Extreme Gradient Boosting.** XGB is an ensemble learning method capable of capturing complex, non-linear relationships in network traffic data. It provides high predictive performance and is effective for structured feature-based classification.
- **Convolutional Neural Networks.** CNN is a deep learning model that can automatically extract hierarchical features from input data. In DDoS detection, CNN is often used to learn spatial patterns in traffic representations, enabling high detection accuracy. However, its computational complexity may limit its suitability for real-time scenarios compared to lightweight models.
- **Light Gradient Boosting Machine.** LightGBM is a gradient boosting framework designed for efficiency and scalability. By employing histogram-based learning and leaf-wise tree growth, it achieves faster training and lower memory consumption while maintaining high detection accuracy.
- **CatBoost.** CatBoost is a gradient boosting algorithm that incorporates advanced techniques for handling categorical features and reducing prediction bias. Its robustness and generalization capability make it effective for cybersecurity classification problems.

The techniques presented above provide different trade-offs between computational efficiency and predictive performance. In this thesis, each model is evaluated independently using the same dataset and experimental settings. Furthermore, an ensemble voting strategy combining LightGBM, XGB, and CatBoost is investigated to leverage the complementary strengths of gradient boosting models and improve overall detection performance.

1.3 Software-Defined Networking and Network Function Virtualization for Adaptive Security

SDN and NFV have introduced greater flexibility and programmability into modern network infrastructures, enabling security services to be deployed and managed more efficiently.

1.3.1 Software-Defined Networking

SDN is a networking paradigm that separates the control plane from the data plane, enabling centralized network management and programmability. In conventional networks, forwarding and control functions are integrated within individual devices, making network-wide policy enforcement difficult and requiring manual configuration on multiple devices. This distributed control structure often increases operational complexity and the likelihood of configuration errors. SDN overcomes these challenges by relocating control functions to a logically centralized entity known as the SDN Controller, which maintains a global view of the network and manages forwarding behavior across the infrastructure.

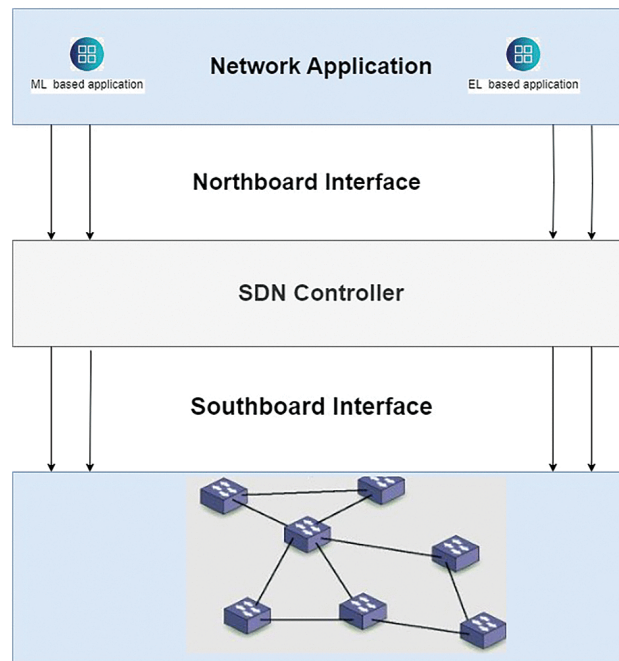


Figure 1.1: *Architecture of SDN [1]*

As illustrated in Figure 1.1, the SDN architecture is organized into three distinct layers. The infrastructure layer forms the foundation of the architecture and consists of forwarding devices responsible for executing forwarding rules defined by the controller. Since forwarding devices no longer perform independent control decisions, cost-effective commercial off-the-shelf (COTS) hardware can be employed instead of specialized proprietary equipment. Above the infrastructure layer lies the control layer, where the SDN Controller operates. Communication between the controller and network devices is achieved through southbound interfaces using

protocols such as OpenFlow, NETCONF, and gNMI. The application layer resides at the top of the architecture and contains network services and management applications, including traffic engineering, load balancing, and security management. These applications interact with the controller through northbound interfaces, commonly exposed through REST-based APIs.

The layered architecture of SDN enables a high degree of network programmability and operational flexibility. By maintaining a centralized view of topology and traffic conditions, the controller can dynamically optimize routing decisions, enforce consistent security policies, and rapidly respond to network events. Consequently, network configurations and policy updates that traditionally require manual intervention can be automated and executed through software interfaces. Such capabilities make SDN particularly suitable for modern data centers, where it is widely adopted to support scalable fabric architectures, network virtualization, and fine-grained security control. The practicality of SDN has been demonstrated by large cloud providers such as Google, Microsoft, and Meta, as well as by open-source platforms including ONOS and OpenDaylight. More recently, artificial intelligence and machine learning techniques have been incorporated into SDN environments to support intelligent traffic analysis, anomaly detection, and adaptive network optimization.

1.3.2 Network Functions Virtualization

NFV is a networking paradigm that decouples network functions from dedicated hardware appliances and implements them as software-based services running on virtualized infrastructure. Instead of relying on proprietary devices, functions such as firewalls, intrusion detection systems, load balancers, and traffic filtering mechanisms can be deployed and managed through software. This approach enables network services to operate independently of the underlying hardware platform.

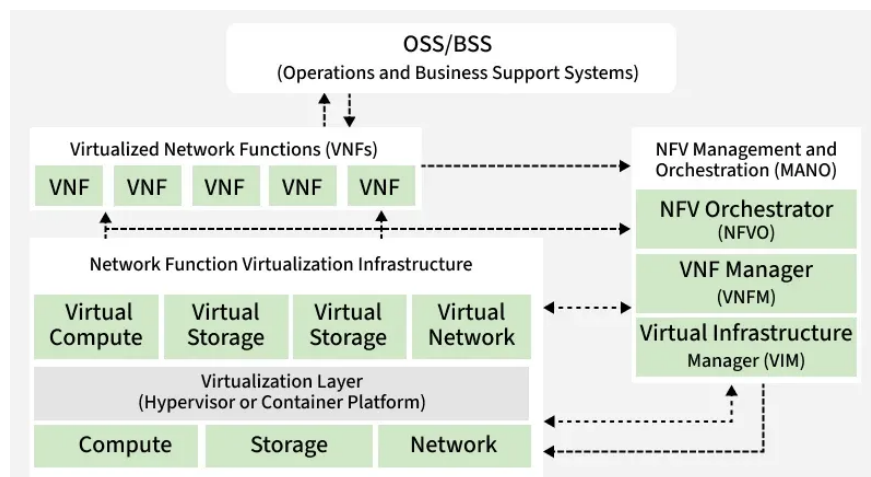


Figure 1.2: *NFV Architecture* [2]

As demonstrated in Figure 1.2, the NFV architecture defined by ETSI consists of three main components: VNFs, NFVI, and MANO. VNFs provide software implementations of

network services, while NFVI supplies the underlying compute, storage, and networking resources required for their execution. MANO coordinates the deployment and lifecycle management of network functions across the infrastructure.

Within the MANO framework, three functional entities are responsible for orchestration and management. The NFVO coordinates network services and resource allocation at the system level. The VNF Manager manages the lifecycle of individual network functions, including instantiation, scaling, updating, and termination. The VIM controls the virtualized resources and provides an abstraction layer between network functions and the underlying infrastructure.

In recent years, NFV has evolved toward cloud-native implementations, where network functions are deployed as containers instead of virtual machines. These functions are commonly referred to as CNFs and are typically orchestrated by Kubernetes-based platforms. This evolution provides the foundation for modern cloud-native networking environments and supports the deployment model adopted in the proposed framework.

1.4 Kubernetes and Containerized Network Functions

Cloud-native technologies have become a key foundation for deploying scalable and flexible network services. Among them, Kubernetes has emerged as the de facto platform for container orchestration, enabling automated deployment, scaling, and management of applications across distributed environments.

- **Kubernetes.** Kubernetes organizes applications into containers running inside pods and provides mechanisms for service discovery, load balancing, resource management, and fault recovery. Its declarative deployment model and orchestration capabilities make it suitable for managing large-scale and dynamic network services.
- **Containerized Network Functions.** CNFs are network functions packaged as containers and deployed on Kubernetes platforms. Compared with traditional virtualized network functions, CNFs offer faster startup times, improved resource efficiency, and easier integration with cloud-native environments. Examples include firewalls, traffic filtering services, intrusion detection systems, and DDoS mitigation functions.
- **Helm Charts.** Helm is a package management tool for Kubernetes that simplifies application deployment through reusable templates known as Helm charts. By defining Kubernetes resources in a structured and reusable manner, Helm enables automated and consistent deployment of CNFs across different clusters and environments.

Together, Kubernetes and CNFs provide a flexible platform for deploying and managing network security functions, supporting adaptive and automated mitigation in modern data center infrastructures.

1.5 The Open Network Automation Platform Closed-loop Orchestration

ONAP [3] is an open-source framework designed to automate the deployment, management, and orchestration of network services. By integrating monitoring, analytics, policy management, and orchestration functions, ONAP enables autonomous network operations and reduces the need for manual intervention.

- **Closed-loop Orchestration.** A closed-loop system continuously monitors network conditions, analyzes collected information, makes decisions based on predefined policies, and automatically executes corrective actions. The feedback generated from these actions is then used to refine subsequent decisions, forming a continuous control cycle.
- **Data Collection and Analytics.** ONAP supports the collection of telemetry data from network services and infrastructure components. These data can be processed by analytics modules to identify anomalies, predict network conditions, or detect security threats.
- **Policy-based Automation.** ONAP incorporates policy-driven decision mechanisms that translate analytical insights into operational actions. This enables automated responses to changing network conditions without requiring direct operator involvement.

By combining monitoring, analytics, policy evaluation, and orchestration within a unified architecture, ONAP provides an effective foundation for implementing AI-driven closed-loop security systems in cloud-native environments.

1.6 Event Streaming and Policy-based Coordination

Modern distributed systems increasingly adopt event-driven architectures to enable scalable and loosely coupled interactions among system components. In such architectures, information is exchanged through events rather than direct service-to-service communication, allowing components to operate independently while maintaining coordinated behavior.

Event streaming platforms such as Apache Kafka provide a reliable mechanism for transporting telemetry data, analytical results, and operational events across distributed environments. By supporting asynchronous communication and real-time data processing, event streaming improves scalability and responsiveness, which are essential for time-sensitive applications such as DDoS defense.

In addition, policy-based coordination enables automated decision-making based on predefined rules and objectives. Analytical outputs can be evaluated against security policies to determine appropriate actions, ensuring consistent and adaptive system behavior. Together, event streaming and policy-driven coordination form a key foundation for implementing closed-loop automation and intelligent network security systems.

1.7 Chapter Summary

This chapter presented the fundamental concepts and technologies underlying the proposed framework. It first discussed the challenges posed by DDoS attacks in modern data center environments and introduced AI-driven detection techniques, including machine learning and deep learning approaches for traffic analysis and anomaly detection. The chapter also reviewed SDN and NFV as enabling technologies for adaptive security, followed by an overview of Kubernetes, containerized network functions, and Helm-based deployment mechanisms in cloud-native environments.

In addition, the chapter introduced ONAP closed-loop orchestration together with event streaming and policy-based coordination concepts, which enable automated monitoring, decision-making, and mitigation workflows. Collectively, these technologies provide the foundation for building an intelligent and adaptive DDoS defense system.

Based on this background, the next chapter reviews existing research on AI-based DDoS detection, mitigation frameworks, and closed-loop security systems, identifying the limitations and research gaps that motivate the proposed approach.

Chapter 2

Related Work

The growing sophistication of DDoS attacks has stimulated extensive research on intelligent detection and adaptive defense mechanisms. Recent studies have investigated a wide range of AI-based techniques for identifying malicious traffic, while programmable network technologies such as SDN and NFV have enabled more flexible mitigation strategies. At the same time, increasing attention has been given to automated and feedback-driven security systems that continuously coordinate monitoring, analysis, and response processes.

This chapter reviews representative studies in these areas, discusses their key contributions and limitations, and identifies the research gaps addressed by the proposed framework.

2.1 AI-Based DDoS Detection Approaches

AI has become a widely adopted approach for DDoS detection due to its ability to learn traffic patterns and identify anomalous behaviors from network data. Compared with traditional signature-based methods, AI-based techniques can detect both known and previously unseen attacks, making them more suitable for dynamic network environments.

Early studies mainly employed machine learning algorithms such as SVM, Random Forests, and KNN for traffic classification. More recently, deep learning models, including CNN and recurrent architectures, have been investigated to automatically extract complex traffic features and improve detection performance.

Ensemble learning techniques such as XGBoost (XGB), LightGBM (GBM), and CatBoost (CB) have also attracted significant attention because of their strong performance on structured network traffic data. Halladay et al. evaluated multiple machine learning and deep learning models on the CIC-DDoS2019 dataset and demonstrated that high detection accuracy could be achieved while reducing computational cost through a compact set of time-based traffic features [4].

Although existing AI-based approaches have achieved impressive detection results, most studies focus primarily on traffic classification. The integration of detection outcomes with adaptive mitigation and continuous feedback mechanisms remains relatively underexplored.

2.2 SDN/NFV-Enabled DDoS Mitigation Frameworks

SDN and NFV has enabled more flexible approaches for DDoS mitigation. SDN provides centralized network control and global visibility, while NFV allows security functions to be deployed and managed as software-based services. Together, these technologies facilitate dynamic traffic management and automated response mechanisms.

Many SDN-based solutions mitigate DDoS attacks by installing flow rules to block, redirect, or rate-limit suspicious traffic once an attack is detected. NFV further enhances flexibility by enabling virtualized security functions, such as firewalls and intrusion detection systems, to be deployed and scaled according to network conditions.

Recent studies have combined SDN and NFV to build adaptive defense frameworks that integrate traffic monitoring, attack detection, and mitigation execution. While these approaches improve network programmability and mitigation flexibility, they often focus primarily on response enforcement, with limited integration between intelligent detection, decision-making, and continuous adaptation.

2.3 Closed-loop and Intent-Based Security Systems

Recent research has increasingly explored closed-loop security systems to improve the automation and adaptability of network defense mechanisms. Unlike conventional approaches that treat monitoring, detection, and mitigation as separate processes, closed-loop architectures continuously connect these functions through a feedback-driven workflow, enabling faster and more coordinated responses to security incidents.

In parallel, intent-based networking and policy-driven management have emerged as promising approaches for simplifying network operations. Instead of manually configuring low-level rules, administrators specify high-level objectives, which are automatically translated into executable policies and enforcement actions. This abstraction improves flexibility and facilitates automated security management in dynamic environments.

Although existing closed-loop and intent-based solutions provide enhanced automation, many of them focus primarily on orchestration and policy enforcement. The integration of AI-driven detection, adaptive mitigation, and continuous feedback within a unified defense framework remains limited, leaving opportunities for more intelligent and responsive DDoS protection systems.

2.4 Research Gaps

The reviewed literature demonstrates significant progress in AI-based DDoS detection, SDN/NFV-enabled mitigation, and automated security management. Existing studies have achieved high detection accuracy through machine learning and deep learning techniques,

while programmable network infrastructures have improved the flexibility of mitigation deployment and traffic control.

However, several limitations remain. First, most research focuses primarily on detection performance, with limited consideration of how detection results can be transformed into timely and effective mitigation actions. Second, detection and mitigation are often implemented as separate components, resulting in delayed responses and reduced adaptability to evolving attack conditions. Finally, although closed-loop and policy-driven security systems have been proposed, the integration of AI-driven detection, adaptive mitigation, and continuous feedback within a unified framework remains insufficiently explored.

These limitations highlight the need for an intelligent closed-loop framework that tightly integrates real-time DDoS detection with adaptive mitigation and feedback-driven decision-making. This research gap motivates the framework proposed in this thesis.

2.5 Chapter Summary

This chapter reviewed representative studies on AI-based DDoS detection, SDN/NFV-enabled mitigation frameworks, and closed-loop security systems. Existing research demonstrates that machine learning and deep learning techniques can achieve high detection accuracy, while programmable network infrastructures provide flexible mechanisms for traffic control and mitigation. Furthermore, recent advances in policy-driven and feedback-based security management have enabled more automated and adaptive defense operations.

Despite these developments, current solutions often focus on individual aspects of the defense process, such as detection, mitigation, or automation, rather than integrating them into a unified system. The limited coupling between AI-driven detection, adaptive mitigation, and continuous feedback reduces the ability of existing frameworks to respond effectively to evolving DDoS attacks in real time.

These observations motivate the development of the AI-driven closed-loop framework proposed in this thesis. The next chapter presents the architecture and design of the proposed system, describing how intelligent detection, adaptive mitigation, and feedback-driven operation are integrated to provide a comprehensive DDoS defense solution for modern data center environments.

Chapter 3

Proposed Method: The CLAD Framework

This chapter presents CLAD, a Closed-loop Learning-based Adaptive DDoS Defense framework for real-time DDoS detection and mitigation in cloud-native data center environments. In this thesis, CLAD refers to the complete closed-loop defense workflow that integrates flow-level telemetry, learning-based traffic classification, intent-based policy decision, CNF-based mitigation, and feedback-driven adaptation. The framework is designed to address the research gaps identified in chapter 2, particularly the limited integration between attack detection, policy-based decision-making, mitigation execution, and post-mitigation feedback in existing DDoS defense approaches.

The central idea of the proposed method is to treat DDoS defense as a continuous closed-loop process rather than a sequence of isolated tasks. Network traffic is continuously monitored and converted into flow-level features. These features are analyzed by AI-based detection models to identify malicious traffic patterns. Once an attack is detected, the detection result is converted into a high-level mitigation intent, which is subsequently translated into executable security policies. Based on these policies, appropriate containerized network functions are selected and activated to mitigate the attack. After mitigation, monitoring information is collected again and fed back into the decision-making process, allowing the system to evaluate whether the response is effective and whether additional actions are required.

By combining AI-driven detection, intent-based policy decision, CNF-based mitigation, and continuous feedback, the proposed framework aims to provide an adaptive and automated defense mechanism for modern data centers. The following sections describe the design objectives, architectural components, core modules, and end-to-end operation workflow of the proposed method.

3.1 Design Objectives and Assumptions

The proposed framework is designed with four main objectives. The first objective is to support accurate and efficient DDoS detection based on flow-level traffic features. Instead of

relying on static signatures or deep packet inspection, the framework analyzes statistical and temporal characteristics of network flows. This design choice reduces monitoring overhead while preserving sufficient behavioral information for distinguishing benign traffic from DDoS attacks.

The second objective is to reduce the gap between attack detection and mitigation execution. In many conventional systems, the detection module only raises alerts, while mitigation is handled manually or by a separate response mechanism. Such separation may introduce operational delay and reduce the effectiveness of defense in fast-changing attack scenarios. To address this issue, the proposed framework directly connects detection results with policy-based mitigation decisions through an automated control workflow.

The third objective is to separate high-level security objectives from low-level enforcement mechanisms. Rather than directly generating device-specific rules from detection outputs, the framework first expresses the desired response as a mitigation intent. This intent-based abstraction allows the system to describe what should be achieved, such as blocking suspicious traffic, rate-limiting excessive flows, or redirecting attack traffic to a scrubber, without binding the decision logic to a specific enforcement technology.

The fourth objective is to support closed-loop adaptation. Mitigation should not be considered the final step of the defense process. After a mitigation action is activated, the system must continue monitoring traffic and system states to determine whether the attack persists, whether the mitigation action is effective, and whether the response should be intensified, adjusted, or withdrawn. This feedback-driven design enables the framework to operate more adaptively under dynamic attack conditions.

The proposed method is based on several assumptions. First, traffic monitoring is performed at the flow level, and extracted features are available to the detection service in near real time. Second, the AI-based detection models are trained offline and deployed as an inference service during system operation. Third, a set of mitigation policies and policy templates is predefined to map detected attack conditions to appropriate response strategies. Fourth, the mitigation functions are implemented as containerized network functions and can be activated or managed through a cloud-native orchestration environment. Finally, the framework assumes that telemetry, detection results, policy decisions, and mitigation events can be exchanged through an event-driven communication mechanism.

3.2 Overall Framework Architecture

The overall architecture of CLAD is illustrated in [Figure 3.1](#). The framework follows a layered closed-loop design that connects traffic monitoring, event processing, AI-based detection, policy decision, mitigation orchestration, and feedback monitoring. For clarity, the architecture can be interpreted through five logical layers: the infrastructure layer, the telemetry and event layer, the intelligence layer, the control and policy layer, and the mitigation layer.

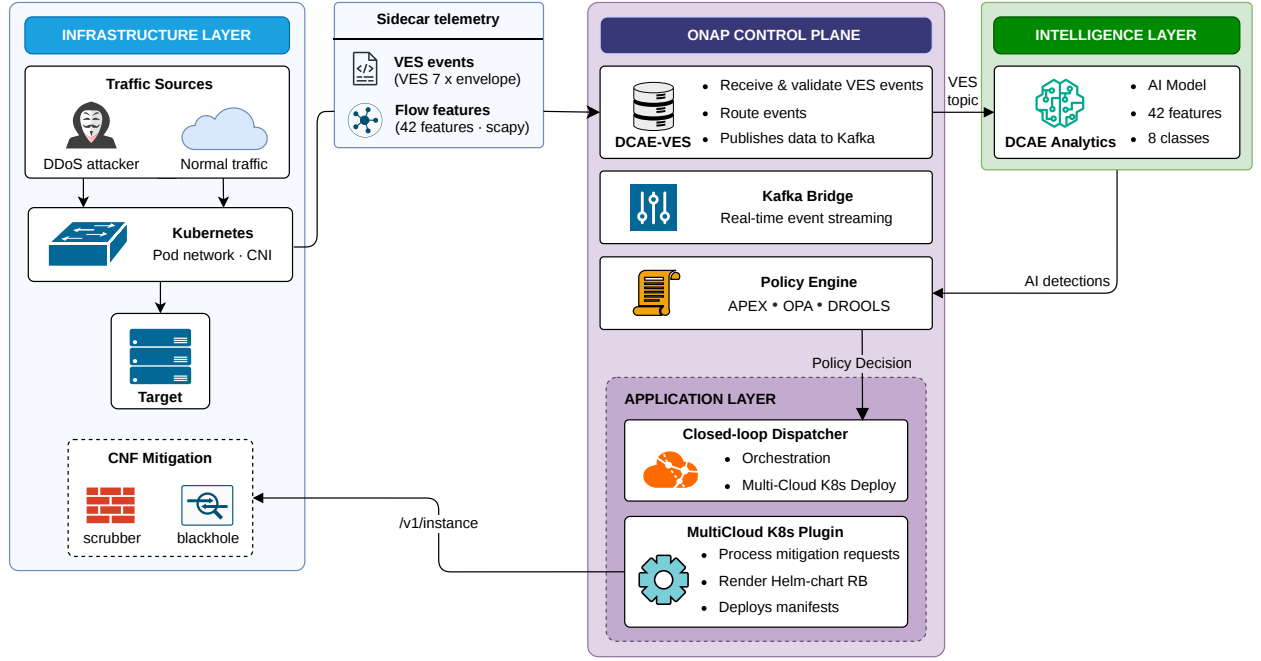


Figure 3.1: Pipeline architecture of the proposed CLAD framework

The infrastructure layer contains the monitored data center environment, including normal traffic sources, potential DDoS attackers, target services, and the underlying cloud-native network. This layer provides the raw traffic from which flow-level features are extracted. Since the proposed framework focuses on scalable data center defense, monitoring is performed at the flow level instead of relying on packet payload inspection. This enables the system to capture traffic behavior while reducing the computational burden associated with deep packet analysis.

The telemetry and event layer is responsible for collecting traffic statistics, generating structured events, and delivering them to other components. Flow-level features are encapsulated into event messages and forwarded through an event streaming mechanism. This design decouples monitoring from analytics and policy decision, allowing each module to operate independently while maintaining real-time coordination through asynchronous event exchange.

The intelligence layer contains the AI-driven detection service. This service receives feature vectors from the event pipeline and classifies the observed traffic into benign or attack categories. The detection output includes the predicted class and may also include additional information such as attack type, confidence score, or severity level. These outputs serve as analytical evidence for the policy decision process.

The control and policy layer transforms detection results into mitigation decisions. It first interprets the detection output as a high-level security intent and then evaluates predefined policies to select an appropriate response strategy. This layer is responsible for deciding whether mitigation should be triggered, what type of mitigation should be applied, and which enforcement component should be activated. By separating policy decision from low-level execution, the framework improves flexibility and allows mitigation strategies to be modified

without redesigning the detection module.

The mitigation layer executes the selected response actions through containerized network functions. Depending on the detected attack characteristics and policy decision, the system may activate functions such as traffic filtering, rate limiting, scrubbing, or blackholing. These functions are deployed and managed in a cloud-native manner, allowing mitigation capabilities to be scaled or adjusted according to attack severity and system conditions.

Table 3.1: *Logical components of the CLAD framework*

Component	Role in the proposed framework
Traffic monitoring service	Collects flow-level traffic statistics from the monitored environment.
Feature extraction module	Converts raw traffic observations into structured statistical and temporal features.
Event streaming mechanism	Transfers telemetry events, detection results, policy decisions, and feedback information among framework components.
AI detection service	Classifies traffic behavior using machine learning and deep learning models.
Intent and policy decision module	Maps detection outputs to high-level mitigation intents and executable response policies.
CNF-based mitigation module	Executes mitigation actions such as filtering, rate limiting, scrubbing, or blackholing.
Feedback monitoring module	Observes post-mitigation traffic and system states to support closed-loop adaptation.

Table 3.1 summarizes the main logical components of the proposed framework. Unlike a purely detection-oriented architecture, the proposed design emphasizes the continuous interaction among analytics, decision-making, enforcement, and feedback. This closed-loop organization is the key mechanism that enables the framework to move from passive attack detection toward automated and adaptive DDoS defense.

3.3 AI-driven Detection Module

The AI-driven detection module acts as the analytical core of the proposed framework. Its role is to identify abnormal or malicious traffic patterns from the flow-level features generated by the monitoring and feature extraction process. The input of this module is a structured feature vector representing the observed network flow or aggregated traffic behavior. The

output is a detection result that indicates whether the traffic is benign or corresponds to a DDoS attack class.

The detection module is designed to support both binary and multi-class classification. In the binary setting, the model distinguishes between benign and malicious traffic. In the multi-class setting, the model further identifies the specific attack category. Multi-class classification is more challenging but more useful for adaptive mitigation, because different attack types may require different response strategies. For example, a volumetric flooding attack may require rate limiting or traffic scrubbing, while a more targeted attack pattern may require stricter filtering or source-based isolation.

To investigate different trade-offs between detection accuracy and computational efficiency, the framework considers several machine learning and deep learning models, including SGD, XGB, LightGBM, CatBoost, and CNN. Lightweight linear models such as SGD are computationally efficient and suitable for low-latency inference, but they may be limited in capturing complex nonlinear relationships in traffic data. Gradient-boosting models such as XGB, LightGBM, and CatBoost are effective for structured tabular features and can model nonlinear interactions among traffic characteristics. Deep learning models such as CNN can learn hierarchical representations, but their effectiveness depends strongly on the input representation and may involve higher computational cost.

In addition to individual classifiers, the framework employs a voting ensemble strategy that combines the outputs of multiple gradient-boosting models. The rationale is that different classifiers may capture different aspects of the traffic distribution. By aggregating their predictions, the ensemble can reduce the effect of individual model bias and improve robustness. In the proposed framework, the ensemble detection result is used as the primary analytical signal for triggering the downstream policy decision process.

The detection module does not directly enforce mitigation actions. Instead, it produces an analytical output that is consumed by the intent-based policy decision module. This separation is important because detection confidence, attack type, and traffic context must be interpreted before a response is selected. In this way, the AI-driven detection module provides evidence for decision-making, while the policy layer determines how the system should react.

3.4 Intent-based Policy Decision Module

The intent-based policy decision module is responsible for transforming AI detection outputs into actionable mitigation decisions. Its input includes the predicted traffic class, attack category, confidence score, and relevant contextual information such as traffic volume, flow rate, affected service, and current mitigation status. Its output is a mitigation policy that can be executed by the mitigation subsystem.

A key design principle of this module is the use of high-level mitigation intents. A mitigation intent describes the desired security objective without specifying all low-level

implementation details. For example, when a high-volume DDoS attack is detected, the intent may be to reduce excessive traffic toward the target service while preserving legitimate access. When suspicious traffic is concentrated from specific sources, the intent may be to block, isolate, or rate-limit those sources. When the attack pattern is uncertain but severe, the intent may be to redirect traffic to a scrubber for additional inspection.

The use of intent abstraction provides three advantages. First, it decouples security objectives from enforcement mechanisms, allowing the same intent to be implemented through different technologies depending on the deployment environment. Second, it makes the decision process easier to understand and maintain, because policies can be expressed in terms of operational goals rather than device-specific commands. Third, it supports adaptive response, because the same detection output can lead to different actions depending on traffic severity, service priority, or current system state.

After an intent is generated, the policy decision module evaluates predefined policy rules to determine the appropriate mitigation action. A policy rule may consider the detected attack type, confidence threshold, traffic intensity, target service, and previous mitigation outcome. For example, if the confidence score exceeds a predefined threshold and the traffic rate is significantly higher than the normal baseline, the policy engine may trigger a mitigation action. If the attack remains active after the initial response, the feedback mechanism may cause the policy module to escalate the response.

The output of this module is an executable mitigation policy. This policy specifies the selected mitigation function, the target traffic or service, the required enforcement action, and the conditions for maintaining or terminating the action. The policy is then passed to the CNF-based mitigation module for execution. Through this design, the framework establishes a structured bridge between AI-based detection and automated mitigation.

3.5 CNF-based Mitigation Module

The CNF-based mitigation module executes the response actions selected by the policy decision module. Its input is an executable mitigation policy, and its output is an activated mitigation function that enforces the required defense action on the affected traffic. The use of containerized network functions enables the framework to deploy and manage security functions in a flexible and cloud-native manner.

The proposed framework considers mitigation functions such as scrubbers, blackhole functions, traffic filters, and rate limiters. A scrubber is used to inspect and clean suspicious traffic before forwarding legitimate traffic to the protected service. A blackhole function may be used when traffic must be dropped aggressively to protect the target system from overload. Filtering functions can block traffic matching specific conditions, while rate limiters can reduce excessive traffic without fully denying access. The selection of the mitigation function depends on the generated policy and the characteristics of the detected attack.

Compared with static hardware-based security appliances, CNF-based mitigation provides

greater deployment flexibility. Mitigation functions can be instantiated, updated, scaled, or terminated according to system requirements. This is particularly useful in cloud-native data center environments, where traffic patterns and service placement may change dynamically. The same mitigation logic can also be packaged and deployed consistently across different clusters or environments.

The CNF-based mitigation module works in coordination with the control and orchestration layer. Once a policy is received, the system determines whether a suitable mitigation function is already available or whether a new CNF instance must be deployed. If the function is already running, the system can immediately update its enforcement rules. If a new instance is required, the orchestration mechanism activates the corresponding CNF and configures traffic redirection or enforcement rules. After the mitigation function becomes active, its operational status and traffic effects are reported back to the feedback monitoring module.

This module is essential for converting analytical decisions into concrete defense actions. While the AI detection module identifies the attack and the policy module decides the response, the CNF-based mitigation module performs the actual enforcement. Together, these modules form the operational core of the proposed closed-loop DDoS defense framework.

3.6 Closed-loop Feedback and Adaptation Mechanism

The closed-loop feedback and adaptation mechanism continuously connects monitoring, detection, decision-making, mitigation, and post-mitigation evaluation. Instead of considering mitigation as the end of the workflow, the proposed framework treats mitigation as an action whose effectiveness must be continuously observed and assessed.

After a mitigation policy is enforced, the monitoring component continues to collect traffic and system information. This feedback may include post-mitigation traffic rate, number of dropped or forwarded flows, CNF running status, target service availability, resource consumption, and persistence of suspicious traffic patterns. These observations are delivered back to the decision-making process through the event pipeline.

The feedback information is used to determine whether the current mitigation action is sufficient. If the attack traffic decreases and the protected service returns to a stable state, the system may maintain the current mitigation policy for a defined period and then gradually relax the restrictions. If the attack persists or changes its behavior, the system may intensify the response by activating additional filtering, increasing rate-limiting strength, or redirecting traffic to a more restrictive mitigation function. If the mitigation function fails or becomes overloaded, the system may trigger redeployment, scaling, or alternative mitigation actions.

This feedback mechanism improves the adaptability of the framework in three ways. First, it allows the system to react to changing attack conditions instead of relying on a one-time decision. Second, it helps reduce unnecessary disruption to legitimate traffic by relaxing

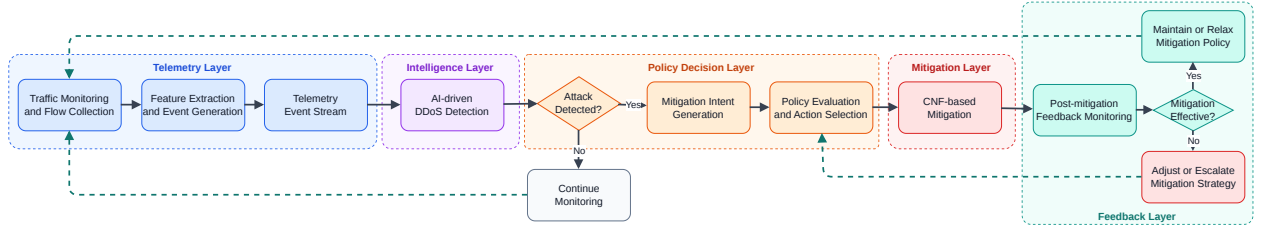


Figure 3.2: *End-to-end closed-loop workflow for AI-driven DDoS detection, policy decision, CNF-based mitigation, and feedback-driven adaptation*

mitigation actions when they are no longer needed. Third, it provides operational evidence for evaluating whether a selected mitigation strategy is effective.

In the proposed framework, feedback-driven adaptation is implemented through the repeated exchange of telemetry, detection, policy, mitigation, and status events. This event-driven design allows the framework components to remain loosely coupled while still participating in a coordinated closed-loop defense process. As a result, the framework can maintain continuous situational awareness and adjust its behavior according to the evolving network state.

3.7 End-to-End Operation Workflow

This section describes the end-to-end operation workflow of the proposed AI-driven closed-loop DDoS defense framework. The workflow explains how the framework continuously transforms network observations into detection results, mitigation decisions, enforcement actions, and feedback-driven adaptation. Unlike a detection-only system, the proposed framework organizes the defense process as a repeated closed-loop cycle in which each stage provides input to the next stage and post-mitigation feedback is used to guide subsequent decisions.

As shown in Figure 3.2, the proposed framework operates through a continuous sequence of monitoring, detection, decision-making, mitigation, and feedback stages. The workflow can be described as follows:

1. **Traffic monitoring and flow collection.** The workflow begins with continuous observation of network traffic in the protected data center environment. Instead of inspecting packet payloads, the monitoring component collects flow-level information such as traffic volume, flow duration, packet rate, communication patterns, and temporal characteristics. This design reduces monitoring overhead and makes the framework more suitable for high-throughput data center networks.
2. **Feature extraction and event generation.** The collected traffic information is transformed into structured statistical and temporal features. These features provide a compact representation of network behavior and serve as the input for the AI-driven detection module. After feature extraction, the resulting feature vectors are encapsulated

into telemetry events and forwarded through the event streaming mechanism. This event-driven design decouples traffic monitoring from downstream analytics and policy decision components.

3. **AI-driven DDoS detection.** The AI-driven detection module receives the extracted feature vectors and performs traffic classification using the deployed detection model. The output of this stage includes the predicted traffic class and, when applicable, the corresponding DDoS attack category. In the proposed framework, the detection module acts as the analytical core that provides evidence for the subsequent policy decision process.
4. **Attack validation and decision triggering.** The detection output is evaluated to determine whether a mitigation response should be triggered. If the observed traffic is classified as benign or does not satisfy the predefined mitigation threshold, the system continues monitoring without activating a response action. If the detection result indicates a confirmed or high-confidence DDoS attack, the workflow proceeds to the mitigation intent generation stage.
5. **Mitigation intent generation.** Once an attack is confirmed, the framework converts the detection result into a high-level mitigation intent. The intent describes the desired security objective, such as blocking malicious traffic, limiting excessive flows, isolating suspicious sources, or redirecting traffic to a scrubber. This abstraction allows the framework to express the response goal without directly binding it to low-level enforcement rules.
6. **Policy evaluation and action selection.** The generated intent is evaluated by the policy decision module. This module considers the detected attack type, confidence level, traffic severity, affected service, and current mitigation state. Based on predefined policy rules, the module selects an appropriate mitigation action and translates the high-level intent into an executable mitigation policy.
7. **CNF-based mitigation execution.** The executable policy is delivered to the mitigation subsystem, where the corresponding containerized network function is activated or updated. Depending on the selected policy, the system may apply filtering, rate limiting, traffic scrubbing, blackholing, or traffic redirection. This stage converts the analytical and policy-level decision into an operational defense action.
8. **Post-mitigation feedback monitoring.** After the mitigation action is enforced, the framework continues collecting traffic and system information. The feedback may include post-mitigation traffic rate, number of dropped or forwarded flows, CNF running status, resource consumption, and target service availability. This information is used to assess whether the mitigation action is effective.

9. **Feedback-based adaptation.** If the mitigation action successfully reduces attack traffic and restores service stability, the system may maintain the mitigation policy for a defined period and then gradually relax it. If the attack persists, changes its pattern, or overwhelms the current mitigation function, the framework may adjust or escalate the response by selecting stronger policies, activating additional mitigation functions, or changing the enforcement strategy.
10. **Continuous closed-loop operation.** The final stage returns the system to continuous monitoring. The workflow is repeated as long as the protected environment remains active. Through this repeated cycle, the framework maintains situational awareness and adapts its response to evolving attack conditions instead of relying on a one-time detection and mitigation decision.

The end-to-end workflow highlights the main methodological contribution of the proposed framework. The framework does not only optimize the accuracy of DDoS detection, but also connects detection results with intent-based decision-making, CNF-based enforcement, and feedback-driven adaptation. This closed-loop organization enables the system to move from passive attack alerting toward automated and adaptive DDoS defense in cloud-native data center environments.

3.8 Chapter Summary

This chapter presented the proposed AI-driven closed-loop framework for real-time DDoS detection and mitigation. The framework is designed to address the limitations of detection-only or manually coordinated defense approaches by integrating monitoring, AI-based analytics, intent-based policy decision, CNF-based mitigation, and feedback-driven adaptation into a unified workflow.

The chapter first introduced the design objectives and assumptions of the framework, emphasizing real-time detection, automated mitigation, intent-based abstraction, and closed-loop adaptation. It then described the overall architecture and explained the roles of the main logical components. The AI-driven detection module was presented as the analytical core that classifies traffic behavior from flow-level features. The intent-based policy decision module was then described as the bridge between detection outputs and executable mitigation policies. The CNF-based mitigation module was introduced as the enforcement layer that activates appropriate defense functions in a cloud-native environment. Finally, the closed-loop feedback mechanism and end-to-end workflow were explained to show how the framework continuously evaluates and adjusts its response.

Overall, the proposed method establishes a complete defense workflow that connects detection, decision-making, mitigation, and feedback within a single adaptive framework. The next chapter presents the implementation environment, dataset configuration, experimental

scenarios, evaluation metrics, and empirical results used to assess the effectiveness of the proposed approach.

Chapter 4

Experiments and Evaluation

This chapter presents the implementation and evaluation of the proposed AI-driven closed-loop framework for real-time DDoS detection and mitigation. The evaluation is organized around two complementary aspects. The first aspect focuses on the performance of the AI-based detection module using benchmark traffic data. The second aspect validates the end-to-end closed-loop workflow, including telemetry collection, AI inference, policy decision, CNF-based mitigation, and feedback monitoring. Through this organization, the chapter evaluates not only the classification capability of the detection models, but also the feasibility of integrating detection and mitigation into an automated cloud-native defense workflow.

4.1 Experimental Testbed and Implementation Environment

The proposed framework was implemented in a cloud-native experimental environment designed to support automated DDoS detection and mitigation. The testbed consists of a Kubernetes-based infrastructure hosting the main framework components, including telemetry collection services, AI analytics modules, policy decision components, event streaming services, and containerized mitigation functions. The control and orchestration workflow follows the ONAP-style closed-loop paradigm, in which monitoring, analytics, policy evaluation, and mitigation execution are coordinated through event-driven interactions.

The experimental environment includes four physical servers divided according to their operational responsibilities. A three-node Kubernetes cluster is used to host the control-plane and decision-making components, while a separate data-plane server is used for traffic generation, attack emulation, and mitigation validation. The hardware configuration of the testbed is summarized in [Table 4.1](#).

Table 4.1: *Hardware configuration of the experimental testbed*

Server		Qty	Resources	Role
Control master	cluster	1	32 cores / 96 GB RAM / 500 GB SSD	Kubernetes master node hosting core control and orchestration services
Control worker	cluster	2	8 cores / 16 GB RAM / 500 GB SSD	Kubernetes worker nodes executing distributed microservices
Data-plane server		1	16 cores / 32 GB RAM / 200 GB SSD	Traffic generation, attack emulation, and mitigation validation environment

All servers run Ubuntu Server 20.04 LTS. The main software components include Kubernetes for container orchestration, Kafka/DMaaP for event streaming, VES-style telemetry collection for event delivery, an AI detection service for traffic classification, a policy decision component for mapping detection results to mitigation actions, and CNF-based mitigation functions such as scrubber and blackhole services. This testbed allows the proposed framework to be evaluated as a complete operational workflow rather than as an isolated detection model.

Table 4.2: *Main software components in the implementation environment*

Component	Function
Kubernetes	Container orchestration platform for deploying analytics, control, and mitigation services
ONAP-style control workflow	Closed-loop coordination for monitoring, policy decision, and mitigation execution
Kafka / DMaaP	Event streaming and asynchronous message exchange among system components
Telemetry collector	Collection and forwarding of traffic and operational events
AI detection service	DDoS detection using machine learning and deep learning models
Policy engine	Mapping of detection results to mitigation intents and executable policies
CNF mitigation services	Scrubber and blackhole functions for attack mitigation
Monitoring services	Collection of post-mitigation feedback and operational status

The data plane is composed of two primary elements: a SONiC-based Spine-Leaf network fabric and an Intrusion Detection System (IDS). The underlying network infrastructure adopts a Spine-Leaf topology in which SONiC switches provide connectivity among workloads distributed across multiple security zones.

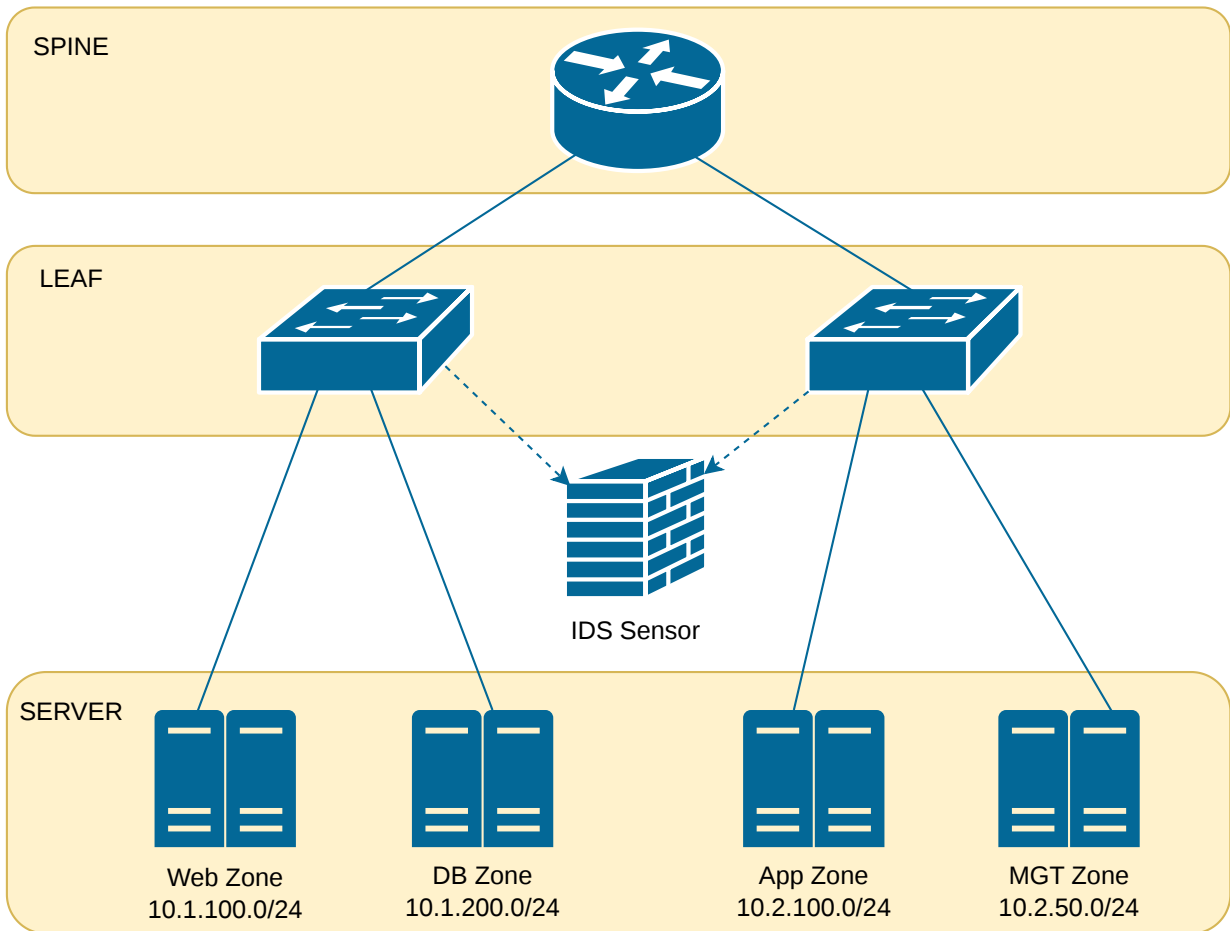


Figure 4.1: *Spine-Leaf topology of the data plane laboratory*

As shown in Figure 4.1, the topology includes a single Spine switch responsible for transit forwarding and two Leaf switches that terminate VLANs and connect end-host workloads. Each security zone is mapped to a dedicated VLAN on the corresponding Leaf switch to provide traffic isolation and facilitate policy enforcement.

Inter-zone communication is governed by a deny-by-default security model, whereby traffic between zones is blocked unless explicitly authorized. The permitted communication patterns are specified in a policy matrix that defines the allowed and prohibited interactions between individual zones. This matrix reflects the intended communication relationships among different application tiers and serves as the basis for enforcing security policies within the data center network.

4.2 Dataset and Preprocessing

The detection models were evaluated using the CIC-DDoS2019 dataset, which is widely used for studying DDoS detection and traffic classification. The dataset contains benign traffic and multiple DDoS attack categories, making it suitable for both binary and multi-class classification tasks. In the binary setting, the objective is to distinguish benign traffic from attack traffic. In the multi-class setting, the objective is to identify the specific DDoS attack category, which provides more informative input for adaptive mitigation.

The preprocessing pipeline converts raw traffic records into structured feature vectors suitable for machine learning. Flow-level statistical and temporal features are used to represent traffic behavior, including information related to packet volume, flow duration, traffic rate, and communication patterns. The use of flow-level features avoids payload inspection and reduces computational overhead, which is consistent with the real-time requirements of data center environments.

Before model training, the dataset is cleaned and transformed into a consistent tabular format. Non-informative or invalid values are removed or handled appropriately. Feature vectors are then prepared for classification and used to evaluate the selected models under the same preprocessing configuration. This consistent preprocessing pipeline ensures that the observed differences in model performance are mainly due to the learning algorithms rather than differences in data preparation.

4.3 Model Training and Configuration

The evaluation considers both individual classifiers and an ensemble classifier. The individual models include SGD, CNN, XGB, LightGBM, and CatBoost. These models represent different learning paradigms and computational trade-offs. SGD is a lightweight linear learning method suitable for efficient inference. CNN represents a deep learning approach capable of learning hierarchical patterns from transformed feature representations. XGB, LightGBM, and CatBoost are gradient-boosting models that are effective for structured tabular traffic features.

In addition to the individual models, a voting ensemble classifier is constructed by combining LightGBM, XGB, and CatBoost. The motivation for using this ensemble is that different boosting models may capture complementary decision boundaries from the same traffic features. By aggregating their predictions, the ensemble can reduce individual model bias and improve classification robustness.

All models are trained and evaluated using the same dataset configuration and preprocessing pipeline. The same set of evaluation metrics is used across all experiments to ensure a consistent comparison. Based on the detection results, the best-performing model is selected as the primary detection engine for integration into the closed-loop mitigation workflow.

4.4 Evaluation Metrics

The detection performance is evaluated using four standard classification metrics: Accuracy, Precision, Recall, and F1-score. Since the detection task includes multi-class classification, the reported Precision, Recall, and F1-score are computed as weighted averages across all classes.

Accuracy measures the overall proportion of correctly classified samples. Precision reflects the reliability of positive predictions, while Recall measures the ability to identify actual attack samples. F1-score is the harmonic mean of Precision and Recall and is particularly useful when the class distribution is imbalanced. These metrics jointly provide a balanced assessment of the detection capability of each model.

For the closed-loop workflow, the evaluation focuses on operational responsiveness. The main measured stages include traffic capture, feature extraction, telemetry generation, AI inference, policy evaluation, CNF deployment, mitigation activation, and post-mitigation feedback monitoring.

4.5 Detection Performance Evaluation

Table 4.3 and Table 4.4 summarize the performance of the evaluated machine learning and deep learning models on the CIC-DDoS2019 dataset for both multi-class and binary classification tasks.

Table 4.3: *Multi-class classification performance of DDoS detection models (%)*

Model	Acc	Prec	Recall	F1
CLAD	97.52	98.18	97.52	97.82
LGBM	97.52	98.18	97.52	97.82
XGB	97.45	98.14	97.45	97.76
CB	97.51	98.14	97.51	97.8
CNN	78.54	70.64	66.51	58.83
SGD	27.52	42.51	27.52	31.39

The results in Table 4.3 reveal a significant performance gap between the gradient-boosting models and the remaining approaches.

The proposed voting ensemble model of CLAD, which combines LightGBM, XGBoost, and CatBoost through weighted voting, achieved the best overall performance with 97.52% accuracy, 98.18% precision, 97.52% recall, and 97.82% F1-score. The voting weights were determined according to the performance of the individual classifiers, resulting in balanced contributions from LightGBM (0.3377), CatBoost (0.3325), and XGBoost (0.3298). This demonstrates that all three boosting models provide valuable predictive information for the final decision process.

Among the standalone classifiers, LightGBM, CatBoost, and XGBoost achieved very similar results, with accuracies exceeding 97% and F1-scores close to 98%. These results confirm the effectiveness of gradient-boosting techniques for learning discriminative patterns from flow-based DDoS traffic features. Although the ensemble model only provides a marginal improvement over the best individual classifier, combining multiple models improves prediction robustness and reduces the impact of model-specific classification errors.

In contrast, the CNN model achieved only 78.54% accuracy and 58.83% F1-score, while SGD obtained 27.52% accuracy and 31.39% F1-score. The significantly lower performance of these models suggests that the selected feature representation is more suitable for boosting-based approaches than for the investigated deep learning and linear learning methods. Based on both classification performance and prediction stability, the voting ensemble model was selected as the primary detection engine integrated into the proposed CLAD framework.

Table 4.4: *Binary classification performance of DDoS detection models (%)*

Model	Acc	Prec	Recall	F1
CLAD	99.99	99.99	99.99	99.99
XGB	99.99	99.99	99.99	99.99
CB	99.97	99.97	99.97	99.97
LGBM	99.98	99.98	99.98	99.98
CNN	99.96	99.96	99.96	99.96
SGD	99.35	99.80	99.35	99.52

Compared with the multi-class, the binary classification results in Table 4.4 show higher performance across all evaluated models, with accuracy values exceeding 99%, indicating that distinguishing between benign and malicious traffic is considerably less challenging than identifying specific attack categories.

The proposed voting ensemble model achieved 99.99% accuracy, 99.99% precision, 99.99% recall, and 99.99% F1-score, matching the performance of the standalone XGBoost classifier. The voting weights were distributed almost equally among LightGBM (0.3342), CatBoost (0.3330), and XGBoost (0.3328), reflecting the similarly strong performance of the three boosting models. CatBoost and LightGBM also produced near-perfect results, with accuracies of 99.96% and 99.95%, respectively.

Although the CNN model achieved a slightly lower accuracy of 99.92%, its overall performance remained highly competitive. SGD obtained the lowest results among the evaluated models, with 99.35% accuracy and 99.52% F1-score, yet still demonstrated strong binary classification capability. Since the performance differences among the boosting-based models are minimal, the binary classification results primarily confirm the effectiveness and stability of the proposed detection approach. Nevertheless, the multi-class evaluation remains more representative of real-world DDoS detection scenarios and therefore serves as the primary criterion for selecting the detection model integrated into the CLAD framework.

4.6 Closed-loop Mitigation Evaluation

To validate the end-to-end operation of the proposed framework, a complete DDoS attack scenario was executed on the experimental testbed. The scenario demonstrates how telemetry collection, AI-based detection, policy-driven decision-making, CNF deployment, and feedback monitoring are coordinated within a unified closed-loop workflow.

Figure 4.2 shows the monitoring interface during the attack generation stage. At this stage, attack traffic is launched toward the target service while the monitoring component collects flow-level traffic statistics.

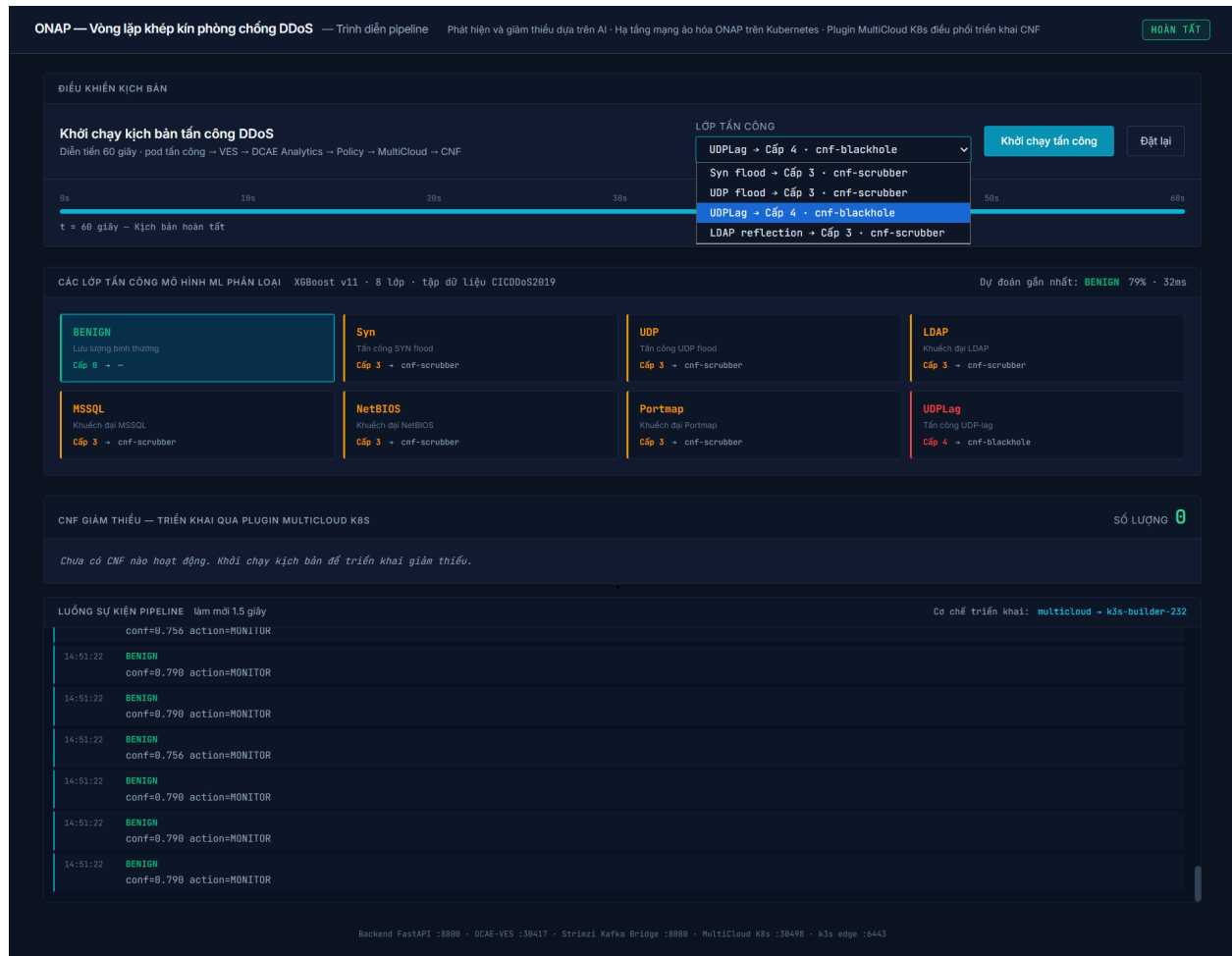
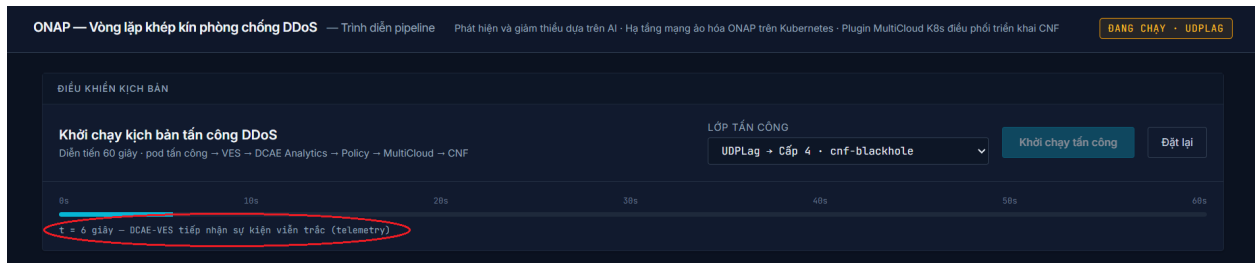
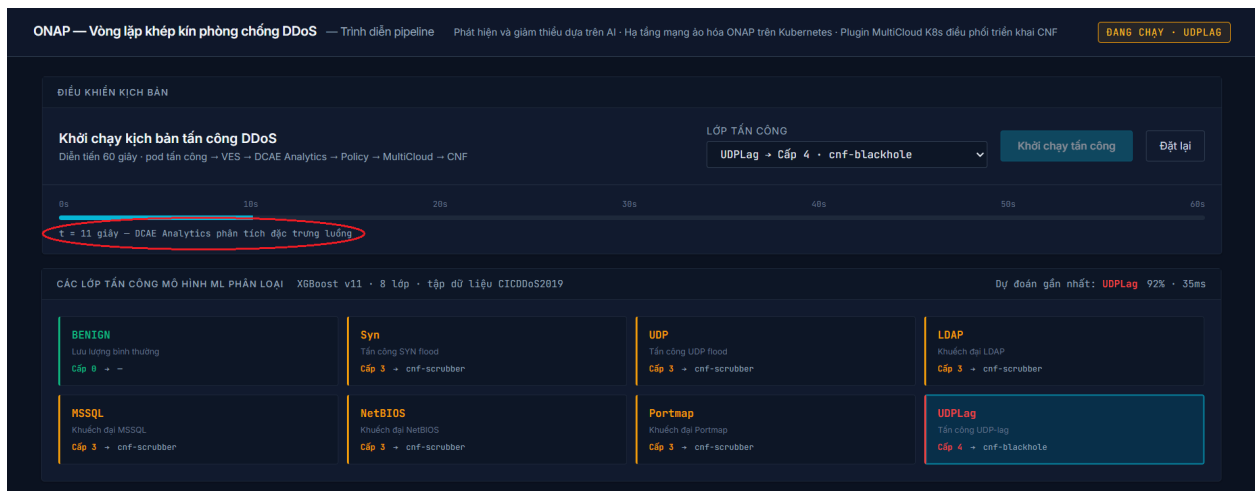


Figure 4.2: Traffic monitoring and attack generation

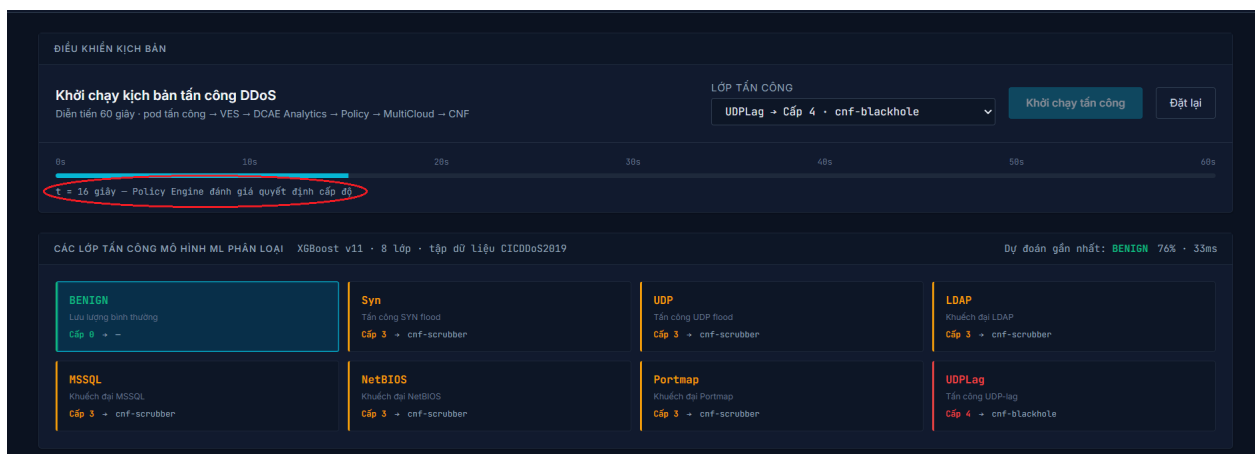
After traffic is captured, the telemetry collection pipeline generates VES-style events and forwards them to the analytics service. This stage confirms that traffic observations can be converted into structured events for downstream processing.

Figure 4.3: *VES collection*

The AI detection service then classifies the extracted traffic features and identifies the corresponding attack category. The detection result is delivered to the policy decision component as the basis for generating a mitigation response.

Figure 4.4: *AI detection and result*

Based on the detection output, the policy engine evaluates predefined mitigation rules and generates an appropriate mitigation intent and response policy. This stage demonstrates the connection between AI analytics and policy-based decision-making.

Figure 4.5: *Policy decision and intent generation*

After the mitigation policy is selected, the dispatcher triggers the deployment of the

required CNF-based mitigation function in the Kubernetes environment. The mitigation service reaches the running state and becomes available to process the attack traffic.

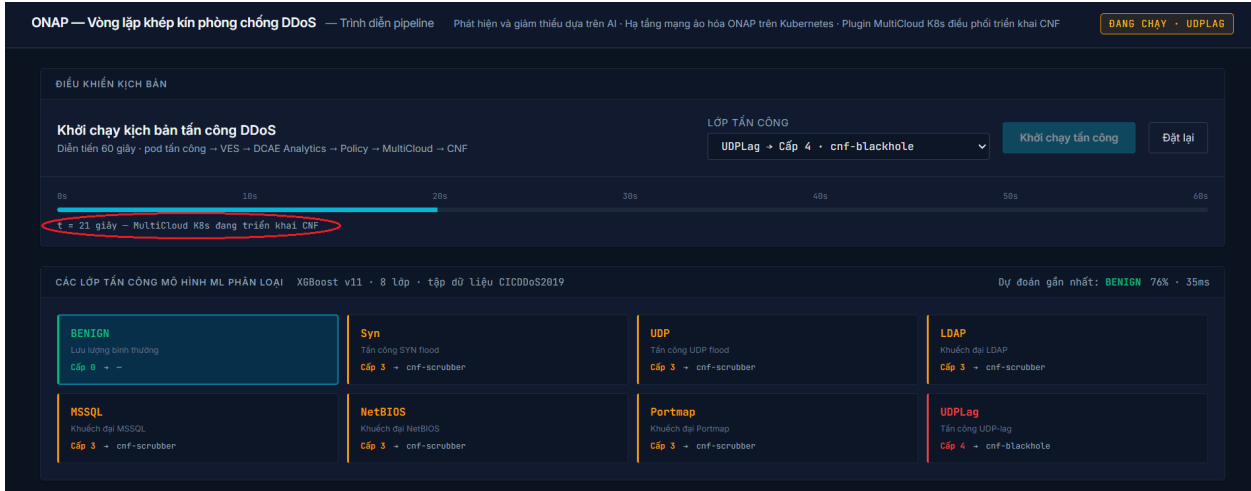


Figure 4.6: CNF deployment

Finally, after mitigation is activated, monitoring services continue collecting operational feedback. This feedback is returned to the control workflow to support closed-loop operation and further decision adjustment if needed.

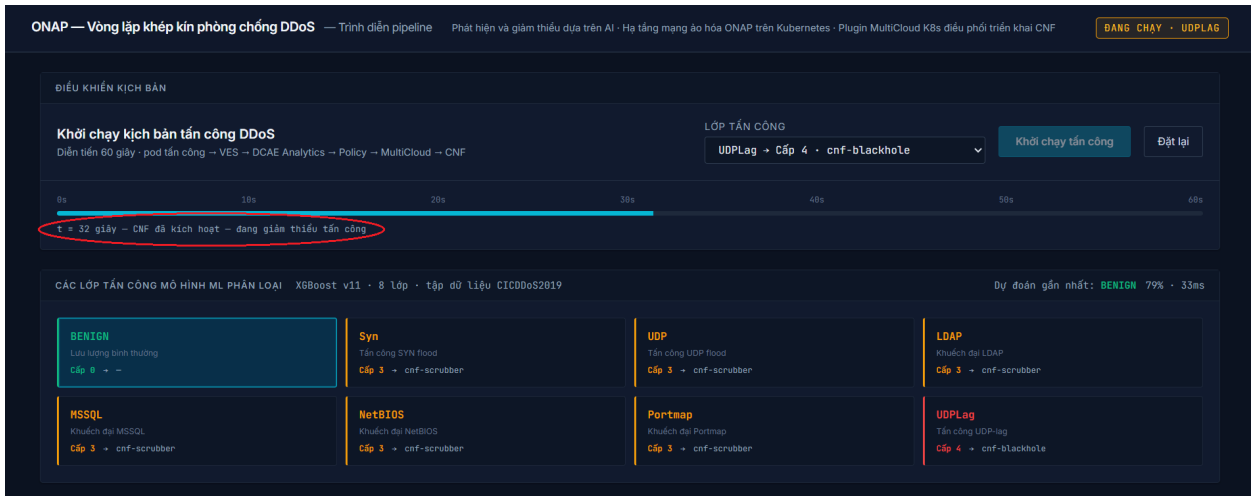


Figure 4.7: Mitigation active and feedback monitoring

The timeline of the closed-loop scenario is summarized in Table 4.5. The full process, from launching the attack to mitigation activation and feedback monitoring, is completed within approximately 60 seconds.

Table 4.5: *End-to-end closed-loop operation scenario*

Time interval	Main activity
0–5 s	Launch DDoS attack and begin traffic capture
5–10 s	Extract flow features and generate telemetry events
10–15 s	Perform AI inference on extracted features
15–20 s	Evaluate mitigation rules and determine response strategy
20–30 s	Trigger mitigation deployment and render Helm chart
30–60 s	Activate mitigation CNF and monitor effectiveness

The results confirm that the proposed framework can coordinate detection and mitigation automatically without manual intervention. The demonstration also shows that the closed-loop architecture successfully integrates AI-based analytics, policy decision, CNF deployment, and feedback monitoring. However, the 60-second end-to-end response time also indicates that CNF activation and orchestration introduce non-negligible delay. In production environments, this latency can be further reduced using pre-deployed mitigation functions, warm standby CNF instances, or hybrid mitigation strategies that combine immediate traffic filtering with dynamic CNF scaling.

4.7 Comparison with Representative State-of-the-Art Methods

Table 4.6 compares the proposed detection models with a representative baseline study on the CIC-DDoS2019 dataset. The comparison includes both multi-class and binary classification settings.

The proposed LightGBM–XGB–CatBoost ensemble achieves the best performance in the multi-class classification task, improving the accuracy from 74.08% and 73.46% in the reported baseline models to 97.52%. The F1-score also increases from 73.42% and 72.57% to 97.82%. These results indicate that the ensemble model is more effective in capturing heterogeneous DDoS traffic patterns than individual boosting models in the referenced baseline.

For binary classification, the evaluated XGB and SGD models achieve accuracies above 99%. However, binary classification is less challenging because it only distinguishes benign traffic from malicious traffic. The multi-class setting provides a more meaningful assessment for the proposed framework, since attack-category information is useful for selecting appropriate mitigation policies.

It should be noted that the comparison with baseline studies is affected by potential differences in preprocessing, feature selection, dataset split, and experimental protocol. Therefore, the comparison should be interpreted as a representative reference comparison rather than a

Table 4.6: *Performance comparison with state-of-the-art methods on the CIC-DDoS2019 dataset (%)*

Method	Model	Acc	Prec	Recall	F1
<i>Multi-class classification</i>					
CLAD (our)	GBM-XGB-CB	97.52	98.18	97.52	97.82
Halladay et al. [4]	XGB	74.08	–	–	73.42
	GBM	73.46	–	–	72.57
<i>Binary classification</i>					
CLAD (our)	GBM-XGB-CB	99.99	99.99	99.99	99.99
Halladay et al. [4]	XGB	98.58	–	–	99.00
	LightGBM	98.44	–	–	98.00

strictly controlled head-to-head benchmark. Nevertheless, the results demonstrate that the selected ensemble model is effective for the detection component of the proposed framework.

4.8 Chapter Summary

This chapter presented the implementation and evaluation of the proposed AI-driven closed-loop DDoS defense framework. The experimental testbed was built on a Kubernetes-based environment with event streaming, AI analytics, policy decision, CNF-based mitigation, and feedback monitoring components. The CIC-DDoS2019 dataset was used to evaluate the detection models under binary and multi-class classification settings.

The detection results show that the LightGBM-XGB-CatBoost voting ensemble achieves the best multi-class performance, with 97.52% accuracy and 97.82% F1-score. This confirms the suitability of gradient-boosting ensemble models for flow-level DDoS traffic classification. The closed-loop evaluation further demonstrates that the proposed framework can automatically progress from traffic monitoring to telemetry generation, AI inference, policy evaluation, CNF deployment, and feedback monitoring within approximately 60 seconds.

Overall, the results support the feasibility of integrating AI-driven detection with policy-based and CNF-based mitigation in a unified closed-loop workflow. At the same time, several observations should be emphasized. First, the reported detection performance depends on the CIC-DDoS2019 dataset and the adopted preprocessing configuration; therefore, additional validation on real-world and cross-domain traffic is necessary. Second, the closed-loop response time demonstrates functional feasibility but may need further optimization for production-scale DDoS defense. Third, CNF-based mitigation provides flexibility and scalability, but cold-start deployment overhead should be reduced through pre-deployed, warm-standby, or hybrid mitigation mechanisms. These observations provide important directions for improving the proposed framework in future work.

Conclusion and Future Works

Main Contributions

This thesis presented an AI-driven closed-loop framework for real-time DDoS detection and mitigation in cloud-native data center environments. The proposed framework integrates traffic monitoring, AI-based attack detection, policy-driven decision-making, dynamic CNF deployment, and continuous feedback into a unified automated workflow. By combining intelligent analytics with closed-loop operation, the framework aims to improve both detection capability and response automation against evolving DDoS attacks.

The framework was implemented on a Kubernetes-based environment integrated with ONAP orchestration components and evaluated using the CIC-DDoS2019 dataset. Experimental results demonstrated the effectiveness of the proposed approach, particularly through the use of an ensemble classifier combining LightGBM, XGB, and CatBoost, which achieved the best overall detection performance. In addition, the experimental demonstration confirmed the successful integration of AI analytics with policy-based mitigation and CNF deployment within a complete closed-loop operation.

The main contributions of this thesis can be summarized as follows:

- Proposed an AI-driven closed-loop framework that integrates DDoS detection, policy-based decision-making, mitigation deployment, and feedback monitoring into a unified architecture.
- Designed and implemented an AI-based detection module using multiple machine learning and deep learning models, including SGD, CNN, XGB, LightGBM, CatBoost, and a voting ensemble classifier.
- Developed a voting ensemble approach combining LightGBM, XGB, and CatBoost, achieving the best multi-class classification detection performance with 97.52% accuracy and 97.82% F1-score on the CIC-DDoS2019 dataset.
- Implemented a cloud-native mitigation workflow based on ONAP orchestration and containerized network functions, enabling automated deployment of mitigation services in response to detected attacks.
- Demonstrated the feasibility of integrating AI analytics into an ONAP-based closed-loop operational workflow for real-time DDoS defense.

Limitations

Despite the promising results obtained in this study, several limitations remain. First, the experimental evaluation was primarily conducted using the CIC-DDoS2019 dataset, which, although widely adopted in the literature, may not fully represent the diversity and complexity of traffic patterns encountered in real-world production environments. Consequently, the generalization capability of the detection models under unseen network conditions requires further investigation.

In addition, the proposed framework relies on predefined policy rules to transform detection outcomes into mitigation decisions. While this approach provides predictable and explainable behavior, it may not always produce the optimal response under highly dynamic attack scenarios. Furthermore, the experimental deployment was performed in a controlled Kubernetes-based environment, which cannot completely capture the scale, heterogeneity, and operational constraints of large-scale cloud data centers. As a result, additional validation is necessary before practical deployment in production networks.

Future Works

Several directions can be explored to further enhance the proposed framework. From the detection perspective, future research may investigate more advanced ensemble learning and deep learning architectures, as well as online learning techniques that enable continuous model adaptation to newly emerging attack patterns. Such capabilities would improve the robustness and long-term effectiveness of AI-based DDoS detection in rapidly evolving network environments.

From the system perspective, the decision-making process can be extended beyond static policy rules by incorporating intelligent policy selection and adaptive mitigation strategies. Techniques such as reinforcement learning may enable the framework to automatically identify the most appropriate mitigation actions according to attack characteristics and network conditions. Furthermore, evaluating the framework in larger multi-cluster or multi-cloud environments and integrating additional cloud-native security functions would provide a more comprehensive assessment of its scalability and operational effectiveness. These enhancements would further strengthen the applicability of AI-driven closed-loop security systems for protecting next-generation cloud-native infrastructures.

Publication

[PMA1] Hanh P. Du, **Anh M. Pham**, Tuan X. Le, Thuy H. Nguyen, Hoa N. Nguyen, “CLAD: Intent-Based Closed-Loop Adaptive DDoS Defense for Cloud-Native Network Management”, submitted to *International Journal of Network Management (Q2-Scopus, SCI)*, 5/2026.

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